

SYDNEY TECHNICAL HIGH SCHOOL

2 Unit Mathematics

Year 11

Assessment Task 1

May 2009

Time Allowed: 70 minutes

Instructions:

- Write using black or blue pen
- Show all necessary working
- Approved calculators may be used
- Start each question on a new page
- Marks may be deducted for careless or badly arranged work

Name: _____ Teacher: _____

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Total

QUESTION 1**Marks**

- a) Evaluate $\frac{1}{\sqrt{5.3^2 - 1.7}}$ correct to 2 decimal places **2**
- b) Express 183479208 in scientific notation correct to 3 significant figures **1**
- c) Simplify $\frac{5}{5y + 20}$ **1**
- d) Given that the surface area of a cone is found using the formula
$$A = \pi rs + \pi r^2$$
find the value of s correct to one decimal place when $A = 150$ and $r = 5$ **2**
- e) Simplify $x^2 + 5x - (x + 2)^2$ **2**

QUESTION 2

- a) Simplify $\sqrt{75} - \sqrt{12}$ **2**
- b) Rationalise the denominator of $\frac{\sqrt{2}}{3\sqrt{5} + 2}$ **2**
- c) Find a and b if $(4 - \sqrt{3})^2 = a - \sqrt{b}$ **2**
- d) State the domain and range of $y = \sqrt{x - 1}$ **2**
- e) Show without sketching that $f(x) = x^2 - x^4$ is an even function **1**

QUESTION 3**Marks****a)** Factorise fully

i) $x^3 + 27$ **1**

ii) $3x^2 - 14x - 5$ **2**

iii) $ab^2 - a + 3b^2 - 3$ **3**

b) Simplify $\frac{x-1}{1-x^3}$ **2**

QUESTION 4**a)** Solve the following

i) $2x^2 + 3x - 7 = 0$ using the general quadratic formula and leaving your answers in exact form. **2**

ii) $\frac{x}{2} - \frac{x+1}{5} = 1$ **2**

iii) $|3y + 2| \geq 5$ **2**

iv) $|x + 1| = 2x + 1$ **3**

QUESTION 5**Marks**

- a) Sketch the following functions on separate number planes. Use a ruler to draw the axes. Label any important points.

i) $y = \frac{-3}{x}$ 2

ii) $3x - 2y - 6 = 0$ 2

iii) $y = (x - 2)^2$ 2

- b) Sketch $x^2 + y^2 = 9$. Is this a function? Why or why not? 2

QUESTION 6

- a) i) Solve simultaneously: 3

$$2x - 3y - 12 = 0$$

$$5x + 2y - 11 = 0$$

- b) i) Sketch $y = x + 1$ and $y = x - 3$ on the same set of axes 2

- ii) Shade the region $y \geq x + 1$ 1

- iii) Write down the inequality which represents the region between the two lines. 1

- c) Margaret Math needed to buy a new calculator as her old one broke. She paid \$26.50 for a new calculator. This price was 7% more than the price of her old calculator. How much did she pay for the old one? 1

QUESTION 7**Marks**

- a) The function $f(x)$ is defined as

2

$$f(x) = \begin{cases} x^3 + 1 & \text{if } x > 2 \\ 2x & \text{if } -1 \leq x \leq 2 \\ 5 & \text{if } x < -1 \end{cases}$$

Find $f(-3) + f(0) - f(4) + f(2)$

- b) If $g(x) = 2x - 1$, find x when $g(x) = 5$

1

- c) Simplify fully $\sqrt{108} \times \sqrt{50}$

2

- d) Simplify $\frac{1}{x} + \frac{2}{x+1}$

2

- e) Evaluate $8.3 \times 10^{15} - 7.1 \times 10^{13}$ and express your answer in scientific notation correct to 2 significant figures.

1**QUESTION 8**

- a) Solve $3^{2x+1} = \frac{1}{27}$

2

- b) Simplify $\frac{3a+6}{4b^3} \div \frac{a^2+a-2}{6b^2-2b}$

3

- c) i) Write down the domain and range of

2

$$y = 2 - \frac{1}{x+1}$$

- ii) Sketch the function showing all main features

2

Teacher's Name:

Student's Name/Nº:

2009 Year 11 2 Unit Ass Task 1 Solutions

Question 1

a) 0.19 ②

① if not rounded off correctly

b) 1.83×10^8 ①

c) $\frac{8}{y(y+4)}$
 $= \frac{1}{y+4}$ ①

d) $A = \pi rs + \pi r^2$
 $150 = \pi \times 5 \times s + \pi \times 5^2$
 $s = 4.5$ ②

① if not rounded

e) $x^2 + 5x - (x+2)^2$
 $x^2 + 5x - x^2 - 4x - 4$ ①
 $x - 4$ ②

Question 2

a) $\sqrt{75} - \sqrt{12}$
 $5\sqrt{3} - 2\sqrt{3}$ ①
 $= 3\sqrt{3}$ ②

b) $\frac{\sqrt{2}}{3\sqrt{5}+2} \times \frac{3\sqrt{5}-2}{3\sqrt{5}-2}$
 $= \frac{3\sqrt{10} - 2\sqrt{2}}{41}$ ②

c) $(4 - \sqrt{3})^2 = a - \sqrt{b}$
 $16 - 8\sqrt{3} + 3 = a - \sqrt{b}$
 $19 - 8\sqrt{3} = a - \sqrt{b}$
 $19 - \sqrt{192} = a - \sqrt{b}$
 $a = 19, b = 192$ ②

d) $y = \sqrt{x-1}$
Domain: $x \geq 1$ ①
Range: $y \geq 0$ ①

e) $f(x)$ even if
 $f(x) = f(-x)$
 $x^2 - x^4 = (-x)^2 - (-x)^4$
 $= x^2 - x^4$ ①
 $\therefore$ Even

Question 3

a) (i) $x^3 + 27$
 $(x+3)(x^2 - 3x + 9)$ ①

(ii) $3x^2 - 14x - 5$
 $\begin{array}{r} + \quad -14 \\ \times \quad -15 \\ \hline -15x - 15 \end{array}$ $\frac{(3x-15)(3x+1)}{3}$

$= (x-5)(3x+1)$ ②

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$$\begin{aligned} \text{ciii) } ab^2 - a + 3b^2 - 3 \\ a(b^2 - 1) + 3(b^2 - 1) \quad (1) \\ (a+3)(b^2 - 1) \quad (2) \\ (a+3)(b-1)(b+1) \quad (3) \end{aligned}$$

$$\begin{aligned} \text{b) } \frac{x-1}{1-x^3} \\ = \frac{x-1}{(1-x)(1+x+x^2)} \quad (1) \\ = \frac{-1}{x^2+x+1} \quad (2) \end{aligned}$$

Question 4

$$\text{a) } 2x^2 + 3x - 7 = 0$$

$$\begin{aligned} x &= \frac{-3 \pm \sqrt{3^2 - 4 \cdot 2 \cdot -7}}{2 \cdot 2} \quad (1) \\ &= \frac{-3 \pm \sqrt{65}}{4} \quad (2) \end{aligned}$$

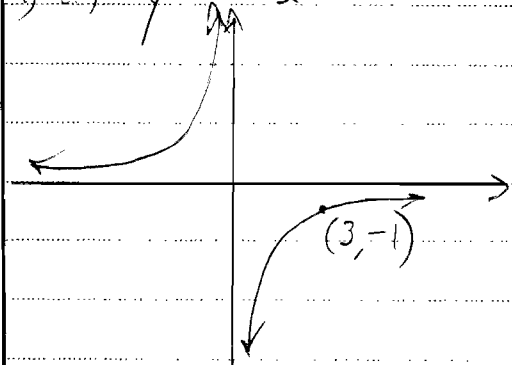
$$\begin{aligned} \text{b) } \frac{x}{2} - \frac{x+1}{5} &= 1 \\ 5x - 2(x+1) &= 10 \quad (1) \\ 3x - 2 &= 12 \\ 3x &= 12 \\ x &= 4 \quad (2) \end{aligned}$$

$$\begin{aligned} \text{ciii) } |3y+2| &\geq 5 \\ 3y+2 &\geq 5 \quad \text{or} \quad 3y+2 \leq -5 \quad (1) \\ 3y &\geq 3 \quad \text{or} \quad 3y \leq -7 \\ y &\geq 1 \quad \text{or} \quad y \leq -\frac{7}{3} \quad (2) \end{aligned}$$

$$\begin{aligned} \text{civ) } |x+1| &= 2x+1 \\ x+1 &= 2x+1, x+1 = -(2x+1) \\ x &= 0 \quad \text{or} \quad x+1 = -2x-1 \\ \uparrow \quad (2) \quad 3x &= -2 \\ x &= -\frac{2}{3} \end{aligned}$$

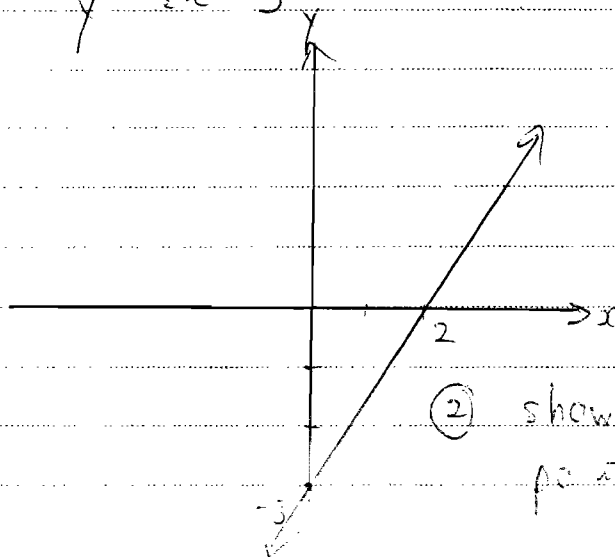
(3) Check! Only $x=0$ Question 5

$$\text{a) i) } y = -\frac{3}{x}$$



(2) showing a pt.

$$\begin{aligned} \text{ii) } 3x - 2y - 6 &= 0 \\ 2y &= 3x - 6 \\ y &= \frac{3}{2}x - 3 \end{aligned}$$

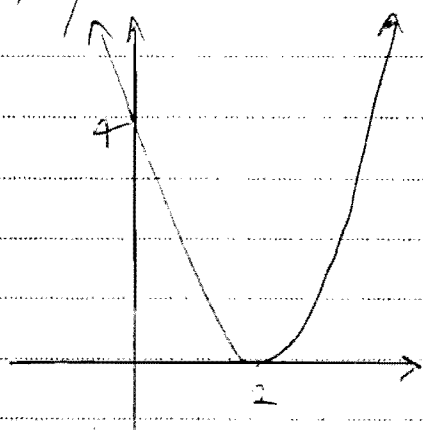


(2) showing 2 points

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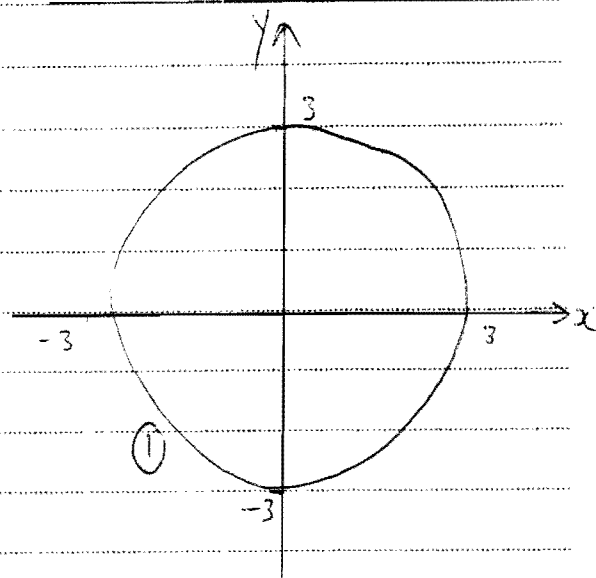
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c(ii) $y = (x-2)^2$



② showing turning point and y intercept

b)



No. Fails vertical line
① test (or similar) ①

Question 6

a) $2x - 3y - 12 = 0$ ①

$5x + 2y - 11 = 0$ ②

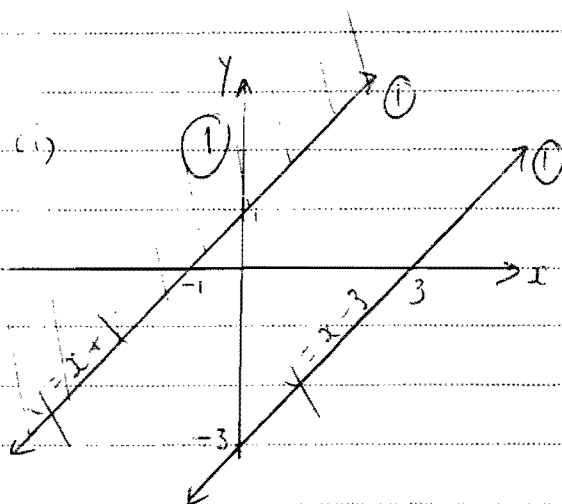
① $\times 2$ + ② $\times 3$

$19x - 57 = 0$ ① mark

$$\underline{x = 3} \quad \underline{y = -2}$$

① ①

b) c)



iii) $x - 3 \leq y \leq x + 1$ ①

or $y \geq x - 3$ and $y \leq x + 1$ ①

d) $\$26.50 \equiv 107\%$
 Old price = 100%
 $\equiv \underline{\$24.77}$ ①

Question 7

a) $f(-3) + f(0) - f(4) + f(2)$

$5 + 0 - 65 + 4$ ①

$\underline{-56}$ ①

b) $g(x) = 2x - 1$

$5 = 2x - 1$

$6 = 2x$

$\underline{x = 3}$ ①

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$$c) \frac{\sqrt{108} \times \sqrt{50}}{6\sqrt{3} \times 5\sqrt{2}} \quad (1)$$

$$= \underline{30\sqrt{6}} \quad (1)$$

$$d) \frac{1}{x} + \frac{2}{x+1} \quad (1)$$

$$\frac{x+1+2x}{x(x+1)} \quad (1)$$

$$\underline{\frac{3x+1}{x(x+1)}} \quad (1)$$

$$e) \underline{8.2 \times 10^{15}} \quad (1)$$

Question 8

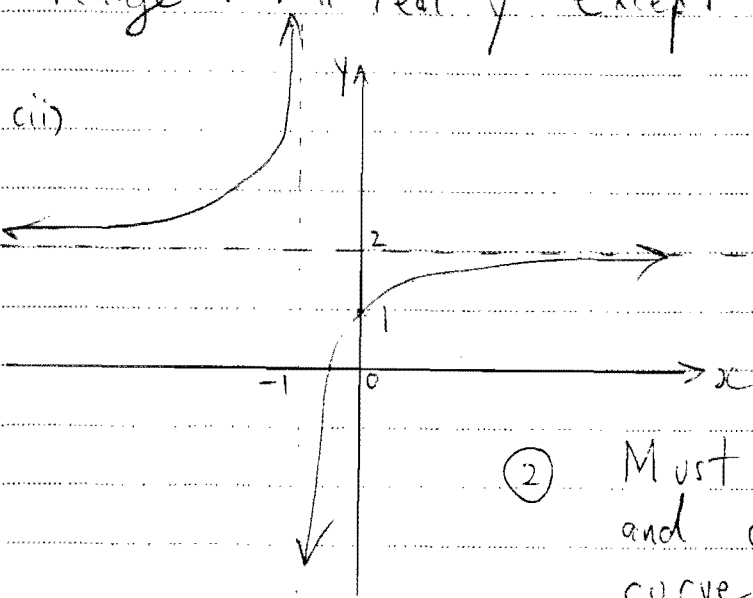
$$a) \begin{aligned} 3^{2x+1} &= \frac{1}{27} \\ 3^{2x+1} &= 3^{-3} \quad (1) \\ 2x+1 &= -3 \\ 2x &= -4 \\ x &= -2 \quad (1) \end{aligned}$$

$$b) \frac{3a+6}{4b^3} \div \frac{a^2+a-2}{6b^2-2b} \quad (2)$$

$$\frac{3(\cancel{a+2})}{4b^3} \times \frac{2b(3b-1)}{\cancel{(a+2)}(a-1)} \quad (2)$$

$$= \underline{\frac{3(3b-1)}{2b^2(a-1)}} \quad (1)$$

$$c) i) y = 2 - \frac{1}{x+1}$$

Domain: All real x except $x = -1$ (1)Range: All real y except $y = 2$ (1)

(2) Must have asymptotes and one point on curve labelled.